

Ender 3 V3 SE + Radxa Zero 3W — Klipper Plan

What Klipper actually changes

Marlin runs planning and step generation on the printer's MCU, inside real-time constraints. Klipper moves trajectory planning to the Radxa and leaves the MCU doing only step execution. Everything below flows from that split — the Radxa isn't making the steppers faster, it's giving the planner room to work.

Level 1 — Software (Thursday)

Stock Klipper stack on the Zero 3W. No new hardware.

- Klipper (jpcurti fork) + Moonraker + Fluidd
- Web control from any device on the network
- Upload straight from the slicer — no SD card shuffle
- Print queue, print history, live telemetry
- Macros: `(PRINT_START FILAMENT=PLA)` runs the whole sequence (heat bed, heat nozzle, home, Z calibrate, mesh, load PA value, go)
- Config changes are a file edit + restart, not a firmware recompile
- printer.cfg under version control — revert a bad change in seconds
- Webcam + timelapse (optional, later)

This alone is most of the value.

Level 2 — Measurement (later)

Add ADXL345. Then find out what the machine actually does:

- Resonance frequency, X and Y → real Input Shaper values, not guesses
- Max volumetric flow — step PLA up until under-extrusion starts, record the number
- Acceleration vs. quality — 1000 / 2500 / 4000 / 6000 mm/s², print the same part at each, look at the results
- Thermal behavior — heatup times, overshoot, stability, fan response

Level 3 — Tuning

Pressure Advance, Input Shaper, acceleration, velocity, bed mesh, Z offset, PID, cooling. Using the Level 2 numbers instead of forum defaults.

Level 4 — Hardware (only if Level 3 hits a wall)

Hotend, part cooling, linear rail, extruder, load-cell sensing. Do this after you know which limit you're actually hitting.

What it does and doesn't do

Claim	Reality
Pressure Advance	Compensates for nozzle pressure change during accel/decel. Does not fix wet filament, bad retraction, or a miscalibrated extruder.
Input Shaper	Reduces resonance-induced ringing. Does not fix belt tension, mechanical play, or worn bearings.
Bed mesh	Compensates for a known bed shape. Does not make the bed mechanically level.
Thermal protection	Marlin already has thermal runaway protection. Klipper gives better <i>visibility</i> , not protection that didn't exist.
Spoolman	Inventory database — spool ID, type, cost, estimated remaining. Does not weigh anything. Real weight needs a load cell.
Jam detection	Not a Klipper feature. Needs added hardware (encoder, filament motion sensor, or load cell) plus custom config.
Power-loss recovery	Possible, but needs real engineering. Position, thermal state, and adhesion after a power cut are all uncertain.
"Beats a KE"	KE has a linear X rail, better hotend, higher rated accel. Some of its advantage is hardware and no amount of firmware fixes that.

Thursday checklist

Before flashing — 5 minutes, do not skip:

- ☐ Read the board revision off the sticker on the mainboard
- ☐ Confirm MCU matches what the jpcurti fork expects
- ☐ Note the display firmware version on first power-on (fork is built against 1.0.6 — newer may need asset re-mapping)

If the board revision doesn't match, stop. Run the printer stock and sort out the right tree before flashing anything.

Flash:

- ☐ Copy `e3v3se_klipper_with_display_C13.bin` to FAT32 microSD
- ☐ **Rename it** — different from the last flashed filename, 8.3 max. Same name = bootloader silently skips = Marlin still on screen
- ☐ Printer OFF → insert card → power on → wait ~2 min
- ☐ Marlin GUI still showing means it didn't take. Rename again, retry

Connect:

- ☐ Printer USB → Radxa (CH340, config uses the by-id path)
- ☐ Deploy staged configs
- ☐ `systemctl restart klipper`
- ☐ Fluidd at `http://radxa-zero3.local:4408`

Then:

- ☐ Print something stock-ish before tuning anything. Find out what the machine does badly before deciding what to fix.
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Firmware staged both places, SHA256 `a193cf05...`:

- *HAL:* `/home/mike/Documents/3d_Printing/firmware/`
- *Radxa:* `/home/pi/firmware/`